Quantification: 876_z76c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	7
interior lacuna count	6
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 6)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1818.83	1904.5	368.311	px^2
major_axis_length_px	lacuna shape	80.949	86.8233	11.0381	px
minor_axis_length_px	lacuna shape	29.7503	29.3475	3.9133	px
aspect_ratio	lacuna shape	2.7475	2.8097	0.4227	unitless
eccentricity	lacuna shape	0.926	0.934	0.0282	unitless
solidity	lacuna shape	0.8569	0.8645	0.0398	unitless
roots_count	headline	9.1667	9	3.8687	count
ring_length_r30_px	headline	590.333	623.5	134.644	px
ring_length_r60_px	reported	1666.5	1787.5	303.315	px
owned_length_px	ownership-dependent	7036.34	6605.6	2300.17	px
edge_count	ownership-dependent	515.333	454.5	189.23	count
mean_edge_length_px	ownership-dependent	13.8066	13.7817	0.7732	px
perimeter_px	normalised	244.001	256.756	33.4039	px
ring_area_r30_px2	normalised	8849.33	9168	778.355	px^2
ring_area_r60_px2	normalised	23250.2	23870.5	1544.15	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0672217	0.074701	0.0162219	px^-1
ring_density_r60	normalised	0.0722589	0.0759454	0.015766	px^-1
roots_per_100px_perimeter	normalised	3.7233	3.5282	1.3213	per 100 px
ring_attached_length_r30_px	network v2	546.833	592.5	154.412	px
ring_attached_length_r60_px	network v2	1455.5	1599.5	453.463	px
ring_length_w_r30_px	network v2	661.785	697.174	152.364	px
ring_length_w_r60_px	network v2	1869.06	2000.31	341.161	px
sholl_crossings_r10	network v2	14.5	14	3.0822	count
sholl_crossings_r20	network v2	18.1667	19	3.1885	count
sholl_crossings_r30	network v2	21	22	5.5136	count
loop_count	loops, report only	71.1667	66.5	31.0704	count
loop_edge_share	loops, report only	0.5407	0.5418	0.0756	share
ring_width_mean_r30_px	width	6.4808	6.6646	0.6048	px

Per field
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measure	value	units
analysed_area_px2	1.03413e+06	px^2
total_skeleton_length_px	65203	px
canalicular_length_density_per_px	0.0630508	px^-1
junction_count	2184	count
junction_density_per_px2	0.00211191	px^-2
lacuna_count	7	count

lacunae_per_px2	6.77e-06	px^-2
skeleton_component_count	667	count
mean_component_length_px	97.7556	px
median_component_length_px	13	px
field_length_density_w_per_px	0.0706444	px^-1
field_density_without_flagged_per_px	0.0642834	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	156	px
unattached_component_count	9	count
canalicular_length_density_attached_per_px	0.0630508	px^-1
field_length_density_with_unattached_per_px	0.0632017	px^-1
loop_count	534	count
loop_edge_share	0.4185	share
edge_count	4894	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.1325	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	53185	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3529	196.715	34.7861	5.655	0.9842	0.5697	-1.4629	16.8161	858.163	True
2	1841	89.0492	27.9031	3.1914	0.9496	0.8121	1.5494	146.123	381.449	False
3	2155	85.0968	33.783	2.5189	0.9178	0.8641	-1.5153	296.21	913.135	False
4	2003	88.5498	30.0304	2.9487	0.9407	0.8649	1.5644	297.345	325.757	False
5	1105	63.6459	23.8313	2.6707	0.9273	0.806	-1.5304	527.835	198.727	False
6	1847	70.5663	34.2892	2.058	0.874	0.8966	-1.5649	538.637	444.666	False
7	1962	88.7858	28.6646	3.0974	0.9464	0.8975	-1.4695	687.621	605.365	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	16	648	1549	5435.72	355	15.3119	3.688
2	16	666	1742	6421.94	424	15.1461	
3	9	447	1298	6789.25	485	13.9985	
4	10	766	1958	9578.03	724	13.2293	6.9134
5	6	602	1894	9857.13	764	12.902	6.4157
6	9	645	1833	5642.86	411	13.7296	6.9445
7	5	416	1274	3928.82	284	13.8339	5.6496

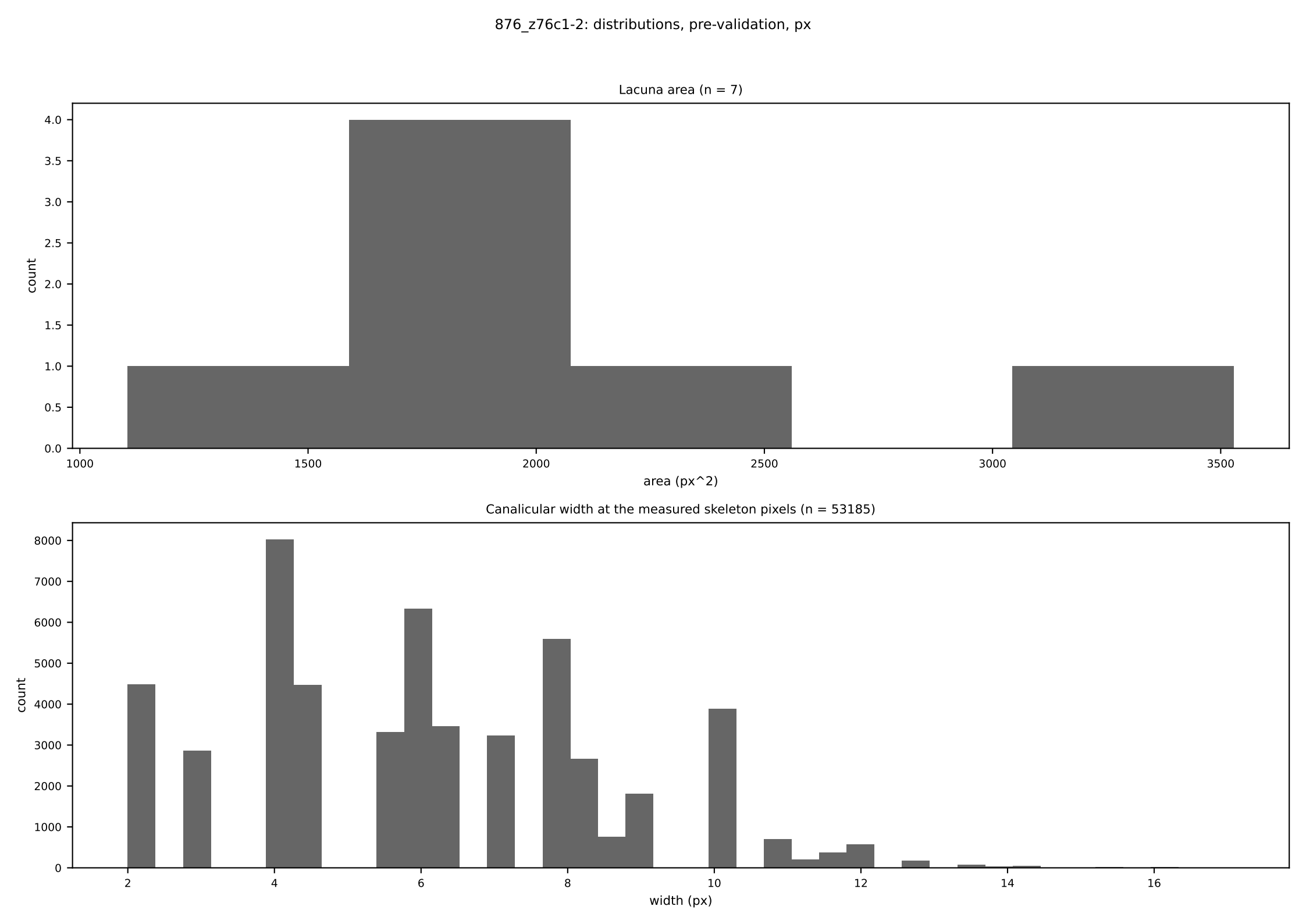
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter

1	756.927	11162	23083	0.6571	0.6039	0.0580541	0.0671057	2.1138
2	269.948	9365	24218	1	1	0.0711159	0.07193	5.9271
3	274.883	9487	24500	1	1	0.0471171	0.0529796	3.2741
4	264.392	9452	24487	1	1	0.0810411	0.0799608	3.7823
5	195.072	7582	20726	1	1	0.0793986	0.0913828	3.0758
6	210.593	8239	22047	1	1	0.0782862	0.0831406	4.2736
7	249.12	8971	23523	1	1	0.0463716	0.0541598	2.0071

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	582	952	722.63	1720.24	17	16	19
2	637	1402	741.529	1933.41	18	21	21
3	402	915	495.99	1435.16	14	12	13
4	733	1838	867.37	2187.52	18	20	29
5	564	1871	672.274	2140.52	14	19	23
6	621	1797	722.073	2067.21	13	18	23
7	324	910	471.475	1450.53	10	19	17


876_z76c1-2: distributions, pre-validation, px
Lacuna area (n = 7)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 53185)
count
width (px)